Niord Public API

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Chapter 1. Introduction

Niord is a system for writing, publishing and archiving Navigational Warnings (NW) and Notices to Mariners (NM). It is developed and run by the Danish Emergency Management Agency.

This document describes the public REST API of Niord: a small, read-only interface that other systems, such as websites, apps and navigation systems, use to fetch the messages that are in force, the publications that can be downloaded, and the areas the messages are filed under. It does not describe the much larger interface that the Niord administration application uses internally; that interface may change at any time and is not meant for other systems.

The public site that Niord comes with is a client of this API. It can be seen at nautiskinformation.brs2.dk, and the Niord Message Presentation document uses it as its example.

The API answers in the format that the Niord Message Model document describes. In the source code, which is published under https://github.com/NiordOrg, the API is the class `ApiRestService` in `be/niord/src/main/java/org/niord/web/api/ApiRestService.java`.

Chapter 2. Niord Public API

The Niord Public API is a read-only REST interface for fetching nautical information from a Niord system: Navigational Warnings (NW), Notices to Mariners (NM), the publications that can be downloaded, and the areas the messages are filed under.

The API needs no login and no key, because it only ever returns public data: messages that are published, or that were published and have since expired or been cancelled, and publications in the categories a system makes public.

2.1. Niord Systems

The API can be used on the following systems.

System	Base URL	Description
Production	niord.brs2.dk	The production system of the Danish Emergency Management Agency.
Staging	niord.staging.brs2.dk	A test system, used to try out new versions before they reach production. It may hold test messages besides copies of real ones, so it is not a source of nautical information.

The public site of each system, nautiskinformation.brs2.dk and nautiskinformation.staging.brs2.dk, passes requests for `/rest/public/v1/` on to its system, so the API can be reached under either host.

2.2. Using the API

All endpoints are below the path `/rest/public/v1` of a system, for example `https://niord.brs2.dk/rest/public/v1/messages`. Some general rules:

- Every endpoint answers `GET` requests (and `HEAD`). There is nothing to create, change or delete.
- The answers are JSON in UTF-8, written as the message model describes: a field without a value is left out, and dates are numbers unless the request asks for text (see the `dateFormat` parameter below).
- A list endpoint returns every match in one answer. There is no paging.
- The API sends no CORS headers (Cross-Origin Resource Sharing), so a script in a web page on another site cannot read its answers in the browser. A website that shows Niord data fetches it from its own server, as the public site does.
- A parameter that can be repeated is given once for each value, for example `mainType=NW&mainType=NM`.
- There is no interactive API page (such as Swagger UI) on the systems. This document, the message model and the XML Schema are the reference.

2.3. Endpoints

The API has seven endpoints:

Endpoint	Description
/rest/public/v1/messages	Searches the messages, by default those in force now.
/rest/public/v1/message/{messageId}	Returns one message by its id or its short id, the id readers know it by, such as `NM-775-26` (see Message in the message model).
/rest/public/v1/publications	Lists the publications that are current, or that were current in a given period.
/rest/public/v1/publication/{publicationId}	Returns one publication by its id.
/rest/public/v1/area/{areaId}	Returns one area by its id or its MRN (Maritime Resource Name, a globally unique name such as `urn:mrn:iho:country:dk`; see Area in the message model), with its parent areas.
/rest/public/v1/area/{areaId}/sub-areas	Lists the areas directly below an area.
/rest/public/v1/xsd/{schemaFile}	Returns the XML Schema of the message model.

2.3.1. Common parameters

Most endpoints take some of the following parameters. Each endpoint's table says which.

Parameter	Type	Default	Description
lang	String	none	A two-letter ISO 639-1 language code, for example `en`. With it, every list of localized texts (`descs`) in the answer holds only the entry in that language. Where a message, area or publication has no text in that language, the list holds the first entry it has instead, with that entry's own language code, so a client can tell. The names in the GeoJSON properties of a message are not filtered; the `language` property of each feature says which language was asked for. Without `lang`, every language is returned.
externalize	Boolean	`true`	Whether to turn the links in the answer into full addresses on the system's own host. For a message, this covers the attachment paths and the links in the HTML of the message parts that point at the message's own files. Without it, an attachment path is the path of the file in the system's repository (`messages/...`), served under `/rest/repo/file/`. For a publication, this covers the `link` of each description that is not already a full address.

Parameter	Type	Default	Description
dateFormat	String	`UNIX_EPOCH`	How dates are written: `UNIX_EPOCH`, a number of milliseconds since 1 January 1970 UTC, or `ISO_8601`, a text such as `"2026-08-18T08:13:26.000+00:00"`. The ISO form is there for clients generated from a schema, which often cannot read the numbers.

2.3.2. /rest/public/v1/messages

Searches the messages. Without any parameter, it returns the messages in force now (status `PUBLISHED`) in the domains the system publishes by default. A *domain* is the producing system's grouping of message series, for example one domain for the navigational warnings and one for the notices to mariners.

Parameter	Type	Example	Description
lang	String	`en`	See Common parameters.
domain	String, repeatable	`niord-nm`	The domains to return messages from. The messages of every message series of the domains are returned. When every domain named is limited to certain areas, only messages in those areas are returned; the same holds for categories. An unknown domain gives no messages.
messageSeries	String, repeatable	`dma-nm`	The message series to return messages from, in addition to those of the domains named. When neither `domain`, `messageSeries` nor `publication` is given, the domains the system publishes by default are used.
publication	String, repeatable	`baed1efd-7e10-4b22-a2ee-e5ae2a6d6381`	The ids of publications whose messages to return, such as one issue of the weekly Notices to Mariners. The answer then holds the messages of the publications, whether they are still in force, expired or cancelled. The id must be that of a publication that /publication/{publicationId} returns. One more kind is accepted here: an issue of a publication series that has since been withdrawn from the public list. That endpoint answers `404` for a withdrawn issue, but its messages can still be fetched here, so a citation of it keeps working. Any other id is refused (see Error answers).

Parameter	Type	Example	Description
areaId	String, repeatable	`urn:mrn:iho:country:dk`	Only messages in these areas or their sub-areas. An area is named by its id (a number) or its MRN, as for /area/{areaId}. An area that does not exist matches no message. When every domain searched, those named or else those the system publishes by default, is itself limited to areas (see `domain`), the areas of the domains are added to the ones named here, and a message in any of them is returned. `areaId` then widens the answer rather than narrowing it.
mainType	String, repeatable	`NM`	Only messages of this main type, `NW` or `NM`.
wkt	String	`POLYGON((10 54, 13 54, 13 56, 10 56, 10 54))`	Only messages whose geometry crosses or lies inside this area, written as Well-Known Text with longitude before latitude. Messages without a geometry are returned as well, since they may concern the area. A text that is not valid WKT is ignored.
externalize	Boolean	`true`	See Common parameters.
dateFormat	String	`UNIX_EPOCH`	See Common parameters.

The messages come in the order the first domain named keeps its messages in, which is usually the order of the area tree. The search lists only messages written by an editor; a message the system made by itself is left out.

The answer is a list of messages, each one as described in the message model. An example, taken from a test system and shortened (one message, one chart, and the position table cut), with the publication and the source replaced by invented ones:

```
curl 'https://niord.staging.brs2.dk/rest/public/v1/messages?lang=en&mainType=NM'
```


```
[
  {
    "id": "5489fde1-d067-46be-9212-363337046587",
    "created": 1787053480000,
    "updated": 1787115206000,
    "messageSeries": { "seriesId": "dma-nm" },
    "number": 775,
    "shortId": "NM-775-26",
    "mainType": "NM",
    "type": "TEMPORARY_NOTICE",
    "status": "PUBLISHED",
    "areas": [
      {
        "id": 466,
```

```
        "active": true,
        "parent": {
          "id": 453,
          "mrn": "urn:mrn:iho:country:dk",
          "active": true,
          "descs": [ { "lang": "en", "name": "Denmark" } ]
        },
        "descs": [ { "lang": "en", "name": "The Baltic Sea" } ]
      }
    ],
    "charts": [
      { "chartNumber": "189", "internationalNumber": 1336, "active": true, "scale":
100000 }
    ],
    "publishDateFrom": 1787115206000,
    "followUpDate": 1792360800000,
    "references": [ { "messageId": "NM-497-26", "type": "REPETITION_NEW_TIME" } ],
    "originalInformation": true,
    "parts": [
      {
        "indexNo": 0,
        "type": "DETAILS",
        "eventDates": [ { "allDay": false, "fromDate": 1787115206000 } ],
        "geometry": {
          "type": "FeatureCollection",
          "id": "79e867af-9831-42ef-8f6c-7313d46953c0",
          "features": [
            {
              "type": "Feature",
              "id": "9f67fa3f-921f-4c25-b107-98b99ecbea14",
              "geometry": {
                "type": "Polygon",
                "coordinates": [ [ [14.689933333333334, 55.09811666666667],
                                   [14.689733333333333, 55.09968333333333],
                                   [14.688, 55.09961666666667],
                                   [14.687316666666666, 55.09721666666667],
                                   [14.689933333333334, 55.09811666666667] ] ]
              },
              "properties": { "language": "en", "startCoordIndex": 1 }
            }
          ]
        },
        "descs": [
          {
            "lang": "en",
            "subject": "Construction works",
            "details": "<p><strong>In the period 31 August - 16 October 2026,</strong>
construction works will be carried out in an area bounded by a line through
positions 1) - 4).</p>\n<table class=\"position-table\">\n<tbody>\n<tr>\n<td class=
\"pos-index\">1)</td>\n<td class=\"pos-col\">55&deg; 05.887'N - 014&deg;
41.396'E</td>\n</tr>\n...</tbody>\n</table>\n<p><span>Rafts and divers will be
```

```
used.</span></p>\n<p>Mariners are requested to pass with caution.</p>"
          }
        ],
        "hideSubject": true
      }
    ],
    "descs": [
      {
        "lang": "en",
        "title": "Denmark. The Baltic Sea. Rønne. Vesthavnen. Construction works.",
        "vicinity": "Rønne. Vesthavnen.",
        "publication": "<a publication=\"f3a2a685-fe2e-48f7-98de-f6459c6667bb\"
href=\"https://www.example.org/pilot\" target=\"_blank\">Harbour pilot, section
4.</a>",
        "source": "Port authority 18 August 2026"
      }
    ]
  }
]
```


2.3.3. /rest/public/v1/message/{messageId}

Returns one message that is public, that is, published, expired or cancelled.

Parameter	Type	Example	Description
messageId	String, in the path	`NM-775-26`	The id of the message (a UUID) or its short id.
lang	String	`en`	See Common parameters.
externalize	Boolean	`true`	See Common parameters.
dateFormat	String	`UNIX_EPOCH`	See Common parameters.

The answer is one message, in the same form as in the list of /messages. When no public message has the id, the answer is `404` with the text `No message found with ID: <id>`.

```
curl 'https://niord.staging.brs2.dk/rest/public/v1/message/NM-775-26?lang=en'
```

2.3.4. /rest/public/v1/publications

Lists the publications in the categories the system makes public: published issues of publication series, such as the weekly Notices to Mariners, and the other publications in use. Without `from` and `to`, the answer holds the publications that are current now.

Parameter	Type	Example	Description
lang	String	`en`	See Common parameters.

Parameter	Type	Example	Description
from	Number	1735689600000	The start of a period, in milliseconds since 1 January 1970 UTC. Only publications that were still current at or after this time are returned.
to	Number	1767225599999	The end of a period, in milliseconds since 1 January 1970 UTC. Only publications that were current at or before this time are returned. With both `from` and `to`, the answer holds the publications that were current at some time in the period. With only one of them, the period has no limit on the other side.
externalize	Boolean	`true`	See Common parameters.
dateFormat	String	`UNIX_EPOCH`	See Common parameters.

The publications come ordered by category (the category's `priority`, lowest first), then with the newest first, as described in the publication model. An example, taken from a test system and shortened to one publication:

```
curl 'https://niord.staging.brs2.dk/rest/public/v1/publications?lang=en'
```


```
[
  {
    "publicationId": "baed1efd-7e10-4b22-a2ee-e5ae2a6d6381",
    "created": 1789479119721,
    "updated": 1789479184590,
    "category": {
      "categoryId": "dk-dma-weekly-nm-publications",
      "priority": 10,
      "publish": true,
      "descs": [ { "lang": "en", "name": "Notices to Mariners" } ]
    },
    "type": "MESSAGE_REPORT",
    "publishDateFrom": 1789479184379,
    "descs": [
      {
        "lang": "en",
        "title": "NtM Week 38 - 2026",
        "link": "https://niord.staging.brs2.dk/rest/repo/file/publications/baed1efd-
7e10-4b22-a2ee-e5ae2a6d6381/NtM-Week-38-2026.pdf",
        "fileName": "NtM-Week-38-2026.pdf"
      }
    ]
  }
]
```

2.3.5. /rest/public/v1/publication/{publicationId}

Returns one public publication: a published issue of a publication series, or another publication in use, in a category the system makes public.

Parameter	Type	Example	Description
publicationId	String, in the path	`baed1efd-7e10-4b22-a2ee-e5ae2a6d6381`	The id of the publication.
lang	String	`en`	See Common parameters.
externalize	Boolean	`true`	See Common parameters.
dateFormat	String	`UNIX_EPOCH`	See Common parameters.

The answer is one publication, in the same form as in the list of /publications. When no public publication has the id, the answer is `404` with the text `No publication found with ID: <id>`. The answer is the same for an id that never existed and for a publication that is not public, so it reveals nothing about publications that are not public.

2.3.6. /rest/public/v1/area/{areaId}

Returns one area with its parent areas, up to the top of the area tree.

Parameter	Type	Example	Description
areaId	String, in the path	`urn:mrn:iho:country:dk`	The id of the area (a number) or its MRN.
lang	String	`en`	See Common parameters.

The answer is one area as described in the message model. When no area has the id, the answer is `404` with the text `No area found with ID: <id>`. An example from a test system:

```
curl
'https://niord.staging.brs2.dk/rest/public/v1/area/urn:mrn:iho:country:dk?lang=en'
```

```
{ "id": 453, "mrn": "urn:mrn:iho:country:dk", "active": true,
  "descs": [ { "lang": "en", "name": "Denmark" } ] }
```

2.3.7. /rest/public/v1/area/{areaId}/sub-areas

Lists the areas directly below an area, in the order of the area tree. Only areas in use (`"active": true`) are listed, and the areas are given without their parents.

Parameter	Type	Example	Description
areaId	String, in the path	`urn:mrn:iho:country:dk`	The id of the area (a number) or its MRN.
lang	String	`en`	See Common parameters.

When no area has the id, the answer is `404` with the text `No area found with ID: <id>`. An example from a test system, shortened to three areas:

```
curl 'https://niord.staging.brs2.dk/rest/public/v1/area/urn:mrn:iho:country:dk/sub-areas?lang=en'
```


```
[
  { "id": 466, "active": true, "descs": [ { "lang": "en", "name": "The Baltic Sea" } ]
},
  { "id": 473, "active": true, "descs": [ { "lang": "en", "name": "The Sound" } ] },
  { "id": 479, "active": true, "descs": [ { "lang": "en", "name": "The Waters South of Zealand" } ] }
]
```


2.3.8. /rest/public/v1/xsd/{schemaFile}

Returns the XML Schema (XSD) of the message model, made from the same classes as the JSON answers. It describes the structure of a message and the classes it holds, including the GeoJSON classes; the publication model is not part of it.

Parameter	Type	Example	Description
schemaFile	String, in the path	`schema2.xsd`	`schema2.xsd` is the main schema. It imports `schema1.xsd`, which defines the nested lists of numbers that coordinates are made of.

The answer is XML (`application/xml`). Any other file name is refused with `400` (see Error answers).

2.4. Error answers

The API answers an error with one of the following.

Status	Body	When
400	A JSON problem description (`application/problem+json`), for example `{"title":"Bad Request","status":400, "detail":"Failed to convert 'mainType' with value: 'XX'","instance": "/rest/public/v1/messages"}`	A parameter has a value of the wrong kind: a `mainType` other than `NW` or `NM`, a `dateFormat` other than the two allowed, a `from` or `to` that is not a number, or an XSD file name other than `schema1.xsd` and `schema2.xsd`.

Status	Body	When
400	A JSON object with a code and a text, for example `{"code":"PUBLICATION_UNRESOLVABLE", "message":"Unresolvable publication id(s): <id>"}`	The `publication` parameter of /messages names an id it does not accept: one that does not exist, or a publication that is not public (see the parameter). The text is the same whatever the reason. If one id of several is refused, the whole request is.
404	A line of plain text, for example `No message found with ID: <id>`	The message, publication or area asked for does not exist or is not public. The answer's `Content-Type` header says `application/json` all the same, so a client should check the status before it reads the body as JSON.
500	An error description	Something went wrong in the system. Try again later.

2.5. Examples

Each Niord system chooses its own ids for domains, message series and areas. The ids in the parameter tables above are those of the sample data the base product comes with. It has five domains, of which two are published by default: `niord-nw` (navigational warnings, with the message series `dma-nw` and `dma-nw-local`) and `niord-nm` (notices to mariners, with the series `dma-nm`). The other three are `niord-fa` (firing areas), `niord-almanac` and `niord-annex`. These ids work only on a system that runs the sample data: a production system has ids of its own, and a domain it does not know gives an empty answer. A client can learn the ids a system uses from its messages (`messageSeries.seriesId`, and the `id` and `mrn` of the areas), or ask the operator of the system. The examples below therefore write a domain id as `<domainId>`.

Fetch the messages in force in the domains the system publishes by default, in English only:

```
curl 'https://niord.brs2.dk/rest/public/v1/messages?lang=en'
```

Fetch the messages in force in one domain, for example `niord-nm` on a system with the sample data, in English only:

```
curl 'https://niord.brs2.dk/rest/public/v1/messages?lang=en&domain=<domainId>'
```

Fetch the navigational warnings in force, in every language the system has:

```
curl 'https://niord.brs2.dk/rest/public/v1/messages?mainType=NW'
```

Fetch the messages in force in and around a sea area, with dates as text:

```
curl -G 'https://niord.brs2.dk/rest/public/v1/messages' \
     --data-urlencode 'wkt=POLYGON((10 54, 13 54, 13 56, 10 56, 10 54))' \
     --data-urlencode 'lang=en' --data-urlencode 'dateFormat=ISO_8601'
```

Fetch the publications that were current in 2025, then the messages of one of them:

```
curl
'https://niord.brs2.dk/rest/public/v1/publications?lang=en&from=1735689600000&to=17672
25599999'
curl
'https://niord.brs2.dk/rest/public/v1/messages?lang=en&publication=<publicationId>'
```

Chapter 3. Other Public Outputs

Besides the REST API, a Niord system serves a few other things without a login. They are made for Niord's own applications rather than as an interface for other systems, so their form may change more easily than that of the API; they are described here so that a client knows what the links it receives lead to.

3.1. Files of messages and publications

The attachments of a message, the pictures and files its text links to, and the documents of publications are files in the system's repository, served under `/rest/repo/file/`. These are the addresses the `path` of an attachment and the `link` of a publication lead to (see the `externalize` parameter in Common parameters). A client fetches them with a plain `GET`.

3.2. A message as PDF

`/rest/message-reports/message/{messageId}.pdf` returns one message as a PDF document, laid out with the system's standard report. The message is named by its id or its short id, and the parameter `lang` (for example `lang=en`) chooses the language. Only a public message can be fetched; for any other id the answer is an error.

The public site's print button in a message's window opens this address.

3.3. A list of messages as PDF

`/rest/message-reports/report.pdf` returns a list of messages as a PDF document, with at most 1,000 messages. Without a login it is always laid out with the system's *public* report, which a request may name with `report=public`; naming any other report is refused with `403`. The following parameters choose the messages, as on the public site's print button:

Parameter	Description
lang	The language of the document, for example `en`.
mainType	`NW` or `NM`, repeatable. Without it, both.
area	The id of an area, repeatable. Note that this parameter is called `area` here, not `areaId` as in the REST API.
messageSeries	The id of a message series, repeatable. Name the series of the messages wanted: without it, the list holds the published messages of every series, including series the public site does not show.
publication	The id of a publication, to print the messages of that publication.

Example:

```
curl -o messages.pdf \
  'https://niord.brs2.dk/rest/message-
```

```
reports/report.pdf?report=public&lang=en&mainType=NM&messageSeries=dma-nm'
```

Making a PDF takes time, so a system makes only a few at once for callers without a login. When it is busy, it answers `503` at once, and the client should try again shortly.

On the public site's own host (for example nautiskinformation.brs2.dk), both PDF addresses are held to the requests the site itself makes, and any other request is refused with `403`:

- A list accepts only the parameters named above, with letters, digits, `.`, `_` and `-` in their values, and exactly one `report=public`.
- One message needs exactly one parameter, `lang`, whose value is made of the same characters: a request without it is refused there, although the system's own host accepts one. The message id in the path may not hold `..` or an encoded dot, slash, backslash, percent sign, semicolon or null character.

3.4. A map of a message

`/rest/message-map-image/{id}.png` returns a small map of the geometry of a message as a PNG picture, as the administration application shows it beside a message in its lists. The message is named by its `id` (the UUID, not the short id). An unknown id is answered with `404`.

When an editor has given the message a picture of its own instead of the drawn map, the answer is a redirect (`307 Temporary Redirect`) to that file under `/rest/repo/file/`, so a client must follow redirects. Otherwise the map is drawn the first time it is asked for after the message was last changed, which can take a moment, and kept afterwards. For a message without a geometry, and whenever the map cannot be drawn, the picture is a placeholder, and the answer then carries the header `X-Niord-Map-Image: placeholder`.

The map is served from the system's own host only (for example niord.brs2.dk), not through the public site.